LLM FOR FINANCIAL NEWS GENERATION

APRIL 2025

HUGO LATENDRESSE

17-644 - Applied Deep Learning

Carnegie Mellon University

Problem Statement and FNSPID Dataset

Can we train a language model to write text similar to a financial news article?

“The following are the top-rated Industrial stocks according to Validea's Value Investor model based on the published strategy of Benjamin Graham. This deep value methodology screens for stocks that have low P/B and P/E ratios, along with low debt and solid long-term earnings growth.
NOW INC (DNOW) is a small-cap value stock in the Misc. Capital Goods industry. The rating according to our strategy based on Benjamin Graham is 86% based on the firm’s underlying fundamentals and the stock’s valuation. A score of 80% or above typically indicates that the strategy has some interest in the stock and a score above 90% typically indicates strong interest.”

Source: https://www.nasdaq.com/articles/valideas-top-industrial-stocks-based-on-benjamin-graham-12-12-2023

Data Transformations

Step 1: Plain Text

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Step 2: Tokens

[8795, 27864, 29462, 17283, 19779, 24355, 33502, 42700, 9105, 11329, 11927, 20214, 22418, 30196, 14826, 24734, 38052, 11056, 36419, 17575, 9305, 37264, 27165, 11537, 19064, 5938, 22768, 31709, 43965, 23604, 4462, 10105, 43883, 39867, 44887, 48360, 41085, 40829, 23188, 16883, 28653, 2212, 46913, 2853, 11980, 25955, 34382, 2380, 18177, 9870, 35186, 11069, 18310, 35996, 9881, 40443, 15801, 30244, 2575, 42218, 34941, 21737, 38840, 13073, 32810, 45924, 14022, 24412, 16023, 36380, 38969, 42144, 25278, 1210, 24871, 24967, 28544, 19157, 17735, 27723, 46136, 5536, 21026, 38907, 15253, 15125, 4982, 39924, 16814, 29189, 7554, 25059, 43292, 17505, 19324, 4175, 5288, 28554, 4221, 5311, ...

Step 3: Tensors

Sequence Length

Batch size

[[8795, 27864, 29462, 17283],

[19779, 24355, 33502, 42700],

[37926, 17813, 31317, 44926],

[25923, 33615, 25301, 6811],

[49909, 8155, 19481, 4313],

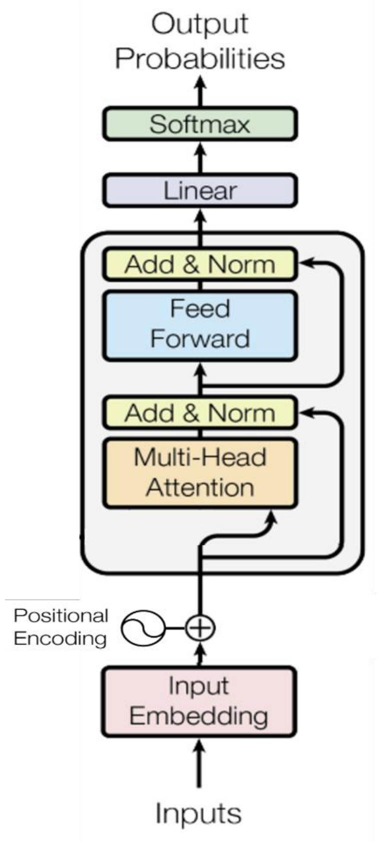
[33353, 21839, 25469, 10582],

[22992, 25365, 17743, 29410],

[31775, 7827, 45208, 1841],

[47746, 9889, 4254, 33222]]

Modular Model Architecture



Source: arxiv.org/pdf/1706.03762.pdf

Modular Model Architecture

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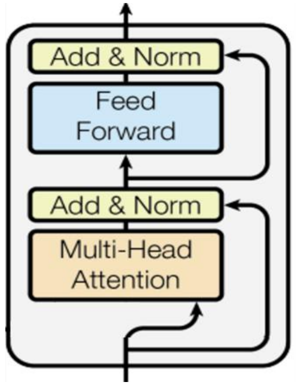
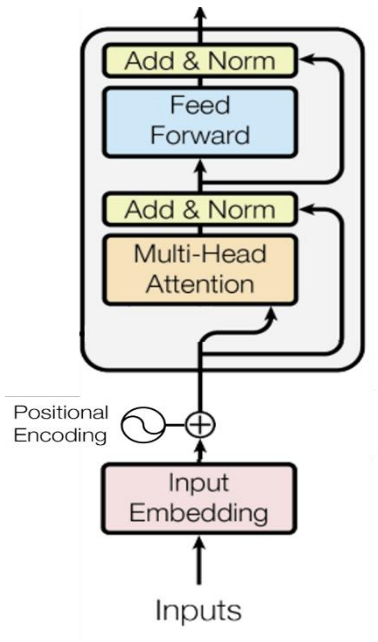
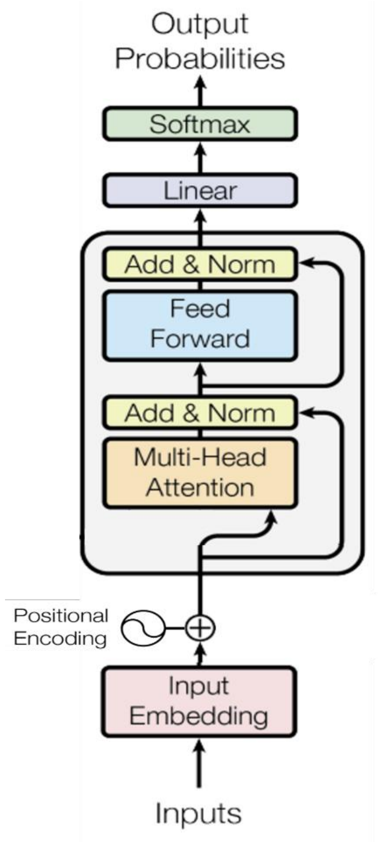


Modular Model Architecture



Modular Model Architecture

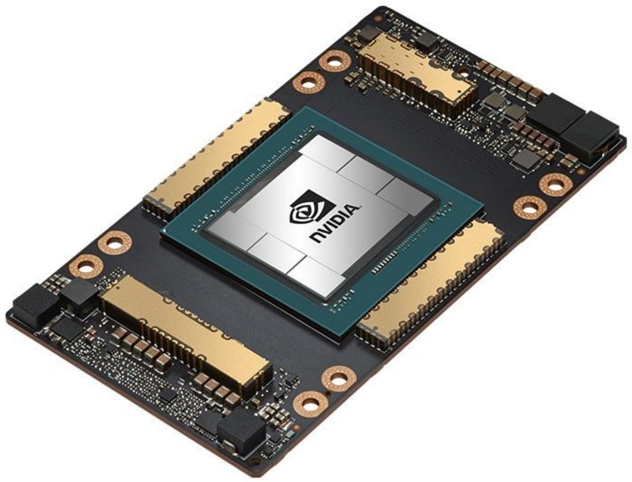


Source: arxiv.org/pdf/1706.03762.pdf

Training on GPU

Intel i7 (My Laptop)
0.06 TLOPS

NVIDIA A100 (Google Colab)
312 TLOPS



First Random Search (10,000 Training Steps)

Weight Decay	Head Count	Block Count	Batch Size	Embedding Size	Learning Rate	Final Validation Loss
0.05	12	6	32	768	0.00010	3.69
0.1	12	12	32	768	0.00010	3.77
0.05	16	12	32	1024	0.00010	3.79
0.1	12	6	32	768	0.00005	3.83
0.1	8	6	32	512	0.00010	4.05
0.1	12	12	8	768	0.00010	4.52
0.1	6	12	8	768	0.00010	4.61
0.05	12	6	32	768	0.00500	6.73

Takeaways

Bigger batch helps

Embeddings should be at least 768

Head count should be at least 12

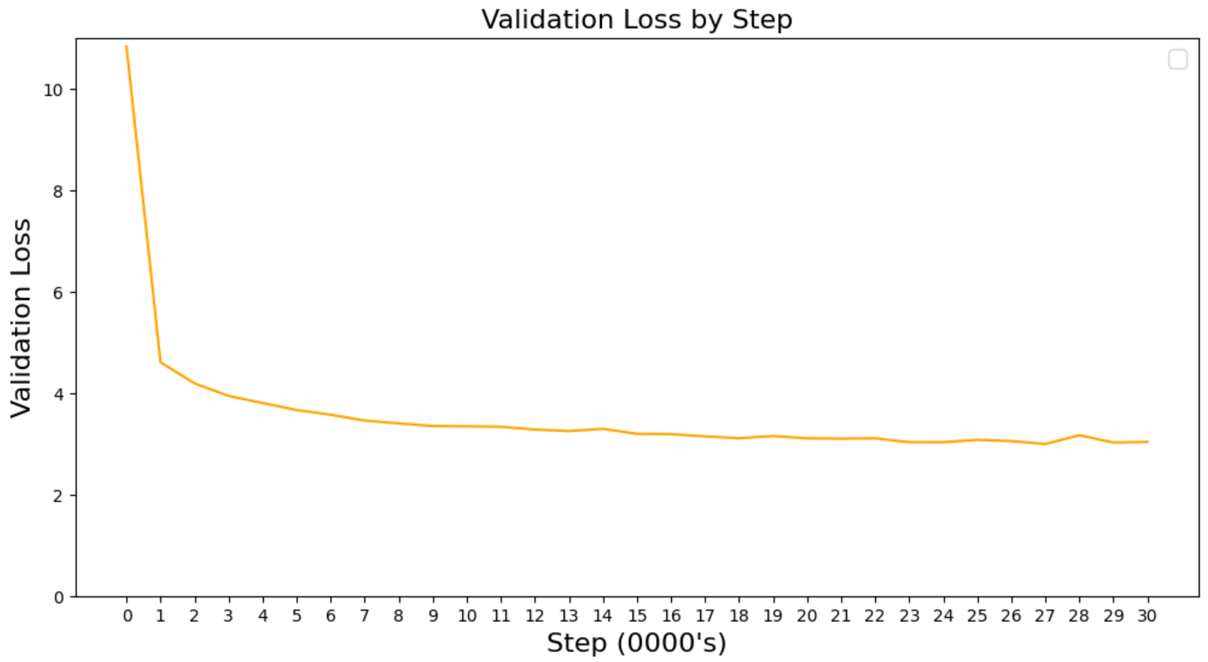
A smaller learning rate is better

Second Random Search (100,000 Training Steps)

Weight Decay	Head Count	Block Count	Batch Size	Embedding Size	Learning Rate	Final Validation Loss
0.1	**12**	**12**	**32**	**768**	**0.00001**	**3.349**
0.05	12	6	32	768	0.00001	3.352
0.05	16	12	32	1024	0.00001	3.425
0.05	12	6	32	768	0.00005	3.537

Final Run

I let the model train until validation loss stops improving



Generation Results

Prompt

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Fintel reports that on December 13, 2023,
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Generation Results

After 10,000 training steps

Fintel reports that on December 13, 2023,**000 shares of A shares, with an average price of $1.00 per share, which includes companies that are not the stock-to-based stock at a Zacks Rank #2 (Buy) stocks. The Zacks Consensus Estimate for the quarter is $1.00 to $3.00, with an average price of $2.30. The stock is expected to report its current price of $1.25 per share is scheduled to be a Zacks Rank #3 (Hold).**

Generation Results

After 10,000 training steps

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Generation Results

After 10,000 training steps

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Generation Results

After 300,000 training steps

Fintel reports that on December 13, 2023, **the company reported a quarterly net loss of $0.02 per share on $1.02 billion in revenue. This compares to the year-ago quarter when earnings came to $0.04 per share on $1.03 billion in revenue.**

What to watch: Shares of Zoom Video Communications are still down about 30% from their all-time high reached on February 12.